Yuan Gao

Ph.D. Candidate & Research Associate — Autonomous Vehicle Systems
Technical University of Munich

Munich, Germany | yuan_avs.gao@tum.de | yuangao-tum.github.io
yuangao-tum | Google Scholar | yuan-gao-tum | ResearchGate



Actively seeking a Ph.D. research internship in foundation models for autonomous driving, embodied AI & robotics — foundation-model-based scenario generation & analysis, agentic frameworks, and vision-language-action (VLA) models. **Available from early 2027** for 3–6 months.

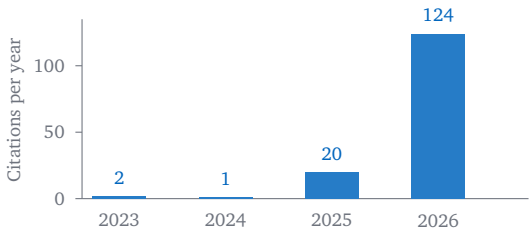
Research Interests

Foundation models (LLMs, VLMs, VLAs, world models) for autonomous driving, embodied AI and robotics — scenario generation and scenario analysis, safety-critical scenario evaluation, agentic frameworks (LLM/VLM agents, multi-agent orchestration, tool use, RAG), vision-language-action models for robot manipulation and navigation, visual question answering and benchmark construction, semantic anomaly detection, agent-level risk assessment; motion planning and optimal control of multi-agent systems.

Research Impact

- **148 citations** total | h-index 5 | i10-index 2
- **124 citations in 2026 alone** — already ∼6× the 2025 total
- **454 GitHub stars** total — 247 on FM-AD-Survey
- **14.2k dataset downloads** on Hugging Face — 10.7k for NuRisk
- **18 research outputs** (13 peer-reviewed) — 11 first-authored

Source: Google Scholar, October 2026; 2026 is year-to-date.



Selected Publications (12 of 18 research outputs)

2027
- **VOLA: Improving Open-World Driving by VLM-Based Semantic Attribute Prediction.**
  IEEE/CVF WACV 2027 • co-author [arXiv]
- **PlugV2V: A Plug-In Vision-Language-Model-Based Semantic Cooperation Agent for Autonomous Driving.**
  IEEE ICRA 2027 • 1st author under review

2026
- **PlannerForge: LLM Agents for Scenario-Based Testing of Motion Planners in Autonomous Driving.**
  EMNLP 2026 (Main Conference) • 1st author [arXiv] [code ★ 11] [project] [dataset]
- **EgoDyn-Bench: Evaluating Ego-Motion Understanding in Vision-Centric Foundation Models for Autonomous Driving.**
  ECCV 2026 • co-author [arXiv] [code ★ 5] [project] [dataset]
- **Target-Bench: Can Video World Models Achieve Mapless Path Planning with Semantic Targets?**
  ECCV 2026 • co-author 🏆 Best Paper, World Models in the Loop workshop ❝ 7 citations [arXiv] [code ★ 56] [project] [dataset]
- **Chat2Scenic: An Iterative RAG-Based Framework for Scenario Generation in Autonomous Driving.**
  IEEE/RSJ IROS 2026 • 1st author (equal contribution) [arXiv] [code ★ 29] [project] [dataset]
- **Can VLMs Unlock Semantic Anomaly Detection? A Framework for Structured Reasoning.**
  IEEE ITSC 2026 • co-author 🏆 Best Student Paper Finalist [arXiv] [project]
- **NuRisk: A Visual Question Answering Dataset for Agent-Level Risk Assessment in Autonomous Driving.**
  IEEE ICRA 2026 • 1st author ❝ 9 citations [arXiv] [code ★ 8] [dataset]
- **Foundation Models in Autonomous Driving: A Survey on Scenario Generation and Scenario Analysis.**
  IEEE Open Journal of Intelligent Transportation Systems • 1st author ❝ 96 citations [DOI] [code ★ 247]

2025
- **From Words to Collisions: LLM-Guided Evaluation and Adversarial Generation of Safety-Critical Driving Scenarios.**
  IEEE ITSC 2025 • 1st author ❝ 22 citations [arXiv] [code ★ 31]

2023
- **H_2 Suboptimal Leader-Follower Consensus Control of Multi-Agent Systems.**
  IFAC World Congress 2023 • 1st author ❝ 3 citations [code ★ 24]
- **H_2 Suboptimal Containment Control of Multi-Agent Systems.**
  American Control Conference (ACC) 2023 • 1st author [IEEE Xplore] [code]

Education

Ph.D. in Mechanical Engineering (Autonomous Vehicle Systems) Oct 2024 – present
Technical University of Munich (TUM), School of Engineering and Design
Professorship of Autonomous Vehicle Systems; advisor: Prof. Dr.-Ing. Johannes Betz.
Research: scenario generation and analysis for autonomous driving based on foundation models.

M.Sc. Development, Production, and Management in Mechanical Engineering Oct 2021 – Aug 2023
Technical University of Munich (TUM)
Grade 1.5 / 1.0 (German scale, 1.0 = best) — passed with distinction (double Master's program).

M.Sc. Mechatronics and Robotics Oct 2019 – Feb 2023
Technical University of Munich (TUM)
Grade 1.2 / 1.0 — passed with high distinction, **top 3% of the cohort**.
Master's thesis at the Chair of Information-oriented Control (Prof. Dr.-Ing. Sandra Hirche, Dr. Junjie Jiao).

M.Sc. studies, Mechatronics and Information Technology Oct 2018 – Sep 2019
Karlsruhe Institute of Technology (KIT)
Continued Master's studies at TUM from Oct 2019.

B.Eng. Mechanical Design, Manufacture, and Automation Sep 2013 – Jun 2017
Hefei University of Technology, China
GPA 3.74/4.3, rank 3/384 — Excellent Graduate honors.
Exchange semester at I-Shou University, Taiwan (Feb – Jun 2015), GPA 4.0/4.0.

Awards & Honors

- **Best Paper Award**, *World Models in the Loop* workshop at ECCV 2026 — Target-Bench.
- **Best Student Paper Finalist**, IEEE ITSC 2026 — SAVANT.
- Double M.Sc. passed with **high distinction** (Grade 1.2 / 1.0, Mechatronics & Robotics — **top 3%** of the cohort) and **distinction** (Grade 1.5 / 1.0), TUM.
- **Excellent Graduate** honors, Hefei University of Technology — undergraduate rank **3 / 384**.
- Perfect GPA (4.0 / 4.0) on exchange semester, I-Shou University, Taiwan.

Research & Professional Experience

Research Associate / Ph.D. Candidate Oct 2024 – present
Professorship of Autonomous Vehicle Systems, TUM
Lead the research line on foundation-model-based scenario generation and analysis for autonomous driving within a project funded by the German Central Innovation Programme for SMEs (ZIM); build and release the accompanying open-source frameworks, benchmarks, and datasets.

Research Assistant Jul 2023 – Nov 2023
Chair of Information-oriented Control, TUM
Distributed optimal control of multi-agent systems. Afterwards (Dec 2023 – Sep 2024), prepared Ph.D. applications and extended the Master's thesis results into a journal submission (IEEE TCNS, under review).

Intern — Healthcare / Medical Robotics Feb 2023 – Jun 2023
ITK Engineering GmbH (Bosch Group)
Internship in the healthcare and medical robotics product area.

Student Research Assistant Mar 2022 – Aug 2022
Chair of Automotive Technology, TUM (Prof. Dr.-Ing. Markus Lienkamp)
Driving-simulator research for automated driving; previously completed a semester thesis at the same chair (May – Dec 2021).

Talks & Presentations

IEEE/RSJ IROS 2026 (Pittsburgh, USA) — presented *Chat2Scenic* in the main conference; talk at the *3rd AI Meets Autonomy: Vision, Language, and Autonomous Systems* workshop. Sep–Oct 2026
IEEE ICRA 2026 (Vienna, Austria) — presented *NuRisk*. Jun 2026
IEEE ITSC 2025 (Gold Coast, Australia) — presented *From Words to Collisions*; workshop talks at the *LLVM-AD* and *Interaction-driven Behavior Prediction & Planning* workshops. Nov 2025
Research visit, Australian Centre for Robotics, University of Sydney. Nov 2025

Open Datasets (Hugging Face, TUM-AVS)

NuRisk — agent-level risk-assessment VQA on BEV scenes (ICRA 2026). 10,739 downloads
Chat2Scenic — Scenic 3.0 scenario database for LLM-based scenario generation (IROS 2026). 1,830 downloads
PlannerForge-Scenarios — 582-scenario CommonRoad corpus for planner testing (EMNLP 2026). 1,021 downloads
munich2drive — 500 surround-camera clips on a real Munich HD map, with ego trajectories. 381 downloads
munich2drive_edgar — 494 clips rendered with TUM's EDGAR vehicle digital twin. 203 downloads

14,174 downloads in total. All-time counts as of October 2026, including the mirrors under `Yuan-avs` and `chat2scenic`.

Academic Service

Conference reviewer: NeurIPS 2026 • ICLR 2026 • AAAI 2027 • CVPR 2026 • WACV 2027 • ICRA 2026, 2025 • IROS 2026 • ITSC 2026, 2025 • IEEE IV 2026, 2025 • IFAC World Congress 2023.
Workshop co-organizer: "AI Meets Control for Resilient and Reliable Robot Planning" (AIM-Ctrl), IEEE/RSJ IROS 2026.
Journal reviewer: IEEE Open Journal of Intelligent Transportation Systems • IEEE Robotics and Automation Letters • IEEE Transactions on Intelligent Transportation Systems.

Teaching (Technical University of Munich)

Teaching Assistant, "F1TENTH: Autonomous Driving Hands-on" (practical course). WS 2024, WS 2025, WS 2026
Teaching Assistant, "Future Mobility Camp" (project-week seminar, practical sessions). WS 2024, WS 2025
Tutor, "F1TENTH: Autonomous Driving Hands-on" (practical course). WS 2024
Tutor, "The Driving Simulator in the Development Process" (practical course, German). SS 2022
Certification: TUM Onboarding Program, elective module "Designing Courses." Dec 2024

Student Supervision (15 theses & projects)

- Vincent Weiskopf, *Automatic Traffic-Rule Video Mining and VQA Benchmarking for Vision–Language Models in Autonomous Driving*, Master's thesis, Jun 2026 – Sep 2026
- Zhiyuan Zhao, *Language-Guided Dynamic Navigation for Autonomous Driving in Closed-Loop Simulation*, semester thesis, Apr 2026 – Oct 2026
- Yang Zhang, *Ethics-Aware Decision Modeling for Autonomous Driving Using Large Vision-Language Models*, Master's thesis, Feb 2026 – Aug 2026
- Yuheng Liu, *CoVLM: A Vision-Language-Model-Based Semantic V2V Cooperation Agent for Autonomous Driving*, Master's thesis, Feb 2026 – Aug 2026
- Chen Li, *Multimodal Retrieval-Augmented Closed-Loop Autonomous Driving Scenario Generation*, Master's thesis, Dec 2025 – Jun 2026
- Joshua Pusch, *A VLM-Based Scenario Analysis of Autonomous Driving*, interdisciplinary project, Sep 2025 – Mar 2026
- Haoyu Wang, *Closing the Loop: Integrating Vision-Language Models for Feedback-Driven ADAS Scenario Generation*, Master's thesis, Sep 2025 – Mar 2026
- Mengchen Cai, *From State Machines to Traffic Safety: Benchmarking Driver Assistance Systems Using Structured Scenario Models*, research internship, Aug 2025 – Feb 2026
- Karmishth Annauth, *Literature Study: Lightweight and Fast-Inference Vision-Language Models for Autonomous Driving*, semester thesis, Aug 2025 – Feb 2026
- Wenting Miao, *Retrieval-Augmented Generation of Simulator-Compatible ADAS Scenarios*, Master's thesis, Jul 2025 – Jan 2026
- Dengyuan Hua, *VLM-based Trajectory Generation: Driving Behavior Diversity*, Master's thesis, Jul 2025 – Jan 2026
- Yung Loon Lim, *Bridging ADAS/ADS Functionality, Regulations, and Scenario-Based Testing: A Comprehensive Literature Review*, semester thesis, Jun 2025 – Dec 2025
- Leonardo Zambrano, *Fine-Tuning Vision-Language Models for Risk Assessment in Autonomous Driving*, Bachelor's thesis, Apr 2025 – Oct 2025
- Sebastian Müller, *Generation of Safety-Critical Scenarios Based on LLMs*, Bachelor's thesis, Mar 2025 – Sep 2025
- Marek Lorenz, *Risk-Aware Scenario Analysis Based on LLMs in Autonomous Driving*, Bachelor's thesis, Oct 2024 – Apr 2025

Skills

Programming	Python, C, C++, C#
ML / Foundation Models	PyTorch, Hugging Face, LLMs & VLMs, agentic AI (LangChain, LangGraph, RAG)
Robotics & Systems	ROS/ROS 2, Linux, Git, CARLA simulation
Modeling & CAD	MATLAB/Simulink, Modelica, SILAB, PLC (CODESYS); AutoCAD, CATIA, Creo, SolidWorks
Languages	Chinese (native) • German (fluent, C1/DSH-2) • English (fluent, C1/IELTS 7.0)